

DRY-RUN GUIDE | 08_quick_sort.py

Source: Python\dsa_solutions\08_quick_sort.py

CORE BEHAVIOR

Orders values using the comparisons, swaps, insertion, partition, or merges shown in this file.

REPRESENTATIVE INPUT

[4,1,3,2].

STEP TRACE

Each pass/partition places values nearer their final positions; completed sequence is nondecreasing.

EXPECTED OUTPUT / RETURN

1 2 3 4 (check source for exact formatting).

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\08_quick_sort.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def partition(numbers, low, high):
005 |     pivot = numbers[high]
006 |     smaller = low
007 |     for index in range(low, high):
008 |         if numbers[index] <= pivot:
009 |             numbers[smaller], numbers[index] = numbers[index], numbers[smaller]
010 |             smaller += 1
011 |     numbers[smaller], numbers[high] = numbers[high], numbers[smaller]
012 |     return smaller
013 |
014 |
015 | def quick_sort(numbers):
016 |     stack = [(0, len(numbers) - 1)]
017 |     while stack:
018 |         low, high = stack.pop()
019 |         if low >= high:
020 |             continue
021 |         pivot_index = partition(numbers, low, high)
022 |         stack.append((low, pivot_index - 1))
023 |         stack.append((pivot_index + 1, high))
024 |
025 |
026 | def main():
027 |     data = list(map(int, sys.stdin.read().split()))
028 |     if not data:
029 |         return
030 |     count = data[0]
031 |     numbers = data[1:1 + count]
032 |     quick_sort(numbers)
033 |     print(*numbers)
034 |
035 |
```

```
036 | if __name__ == "__main__":  
037 |     main()  
038 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.